

Architecture Components

- Components
 - **Client:** End user device (mobile, laptop, TV). Runs the player and picks its own quality.
 - **DNS / GeoDNS:** Points each user at the nearest healthy region.
 - **External Load Balancer:** Terminates TLS, absorbs DDoS traffic, spreads load across gateways.
 - **API Gateway:** Single entry point. Auth, rate limiting, routing, circuit breaking.
 - **Internal Load Balancer:** Distributes requests across service instances.
 - **User Service:** Accounts, profiles, login, authorization.
 - **Subscription Service:** Plans, billing, payments, entitlement checks.
 - **Dashboard Service:** Builds the home screen from recommendations, trending and continue watching.
 - **Search Service:** Search and autocomplete, including misspelled queries.
 - **Watch History Service:** Records progress and serves the continue watching list.
 - **Playback Service:** Creates the session, checks stream limits, returns a signed CDN manifest.
 - **Transcoding Pipeline:** Encodes a master into every resolution and codec, packages it, applies DRM.
 - **Message Queue:** Absorbs write bursts such as progress updates. Failures go to a dead letter queue.
 - **Redis Cache:** Hot metadata, sessions, stream counters. Takes most read load off the databases.
 - **SQL Database:** Accounts, subscriptions, payments. Needs transactions.
 - **Cassandra:** Watch history and resume points. High write rate, eventual consistency.
 - **Elasticsearch:** Inverted index for fuzzy search and autocomplete.
 - **Object Storage:** Master files and encoded segments. Acts as the CDN origin.
 - **CDN:** Caches inside ISP networks that serve the actual video. Almost all bandwidth goes here.
- Playback flow
 - Client requests a playback session.
 - Service checks subscription, region licensing, and concurrent stream limit.
 - Returns resume position and a signed manifest URL pointing at the nearest CDN node.
 - Player pulls segments directly from the CDN and adjusts quality as bandwidth changes.
 - Progress is posted every 30 seconds through the queue into Cassandra.
- Failure handling
 - Recommendations down: fall back to generic trending rows.
 - Cache down: read from the database directly, slower but working.
 - Database node down: a replica serves the request.
 - Region down: GeoDNS reroutes to the next nearest region.
 - CDN node down: player retries against the next closest node.